RAJALAKSHMI ENGINEERING COLLEGE

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N2

Started on	Thursday, 27 August 2026, 11:34 AM
State	Finished
Completed on	Thursday, 27 August 2026, 11:35 AM
Time taken	34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```
 2
 3  void swap(int *a, int *b) {
 4      int temp = *a;
 5      *a = *b;
 6      *b = temp;
 7  }
 8
 9  int partition(int arr[], int low, int high) {
10      int pivot = arr[high];
11      int i = low - 1;
12
13      for (int j = low; j < high; j++) {
14          if (arr[j] < pivot) {
15              i++;
16              swap(&arr[i], &arr[j]);
17          }
18      }
19
20      swap(&arr[i + 1], &arr[high]);
21      return i + 1;
22  }
23
24  void quickSort(int arr[], int low, int high) {
25      if (low < high) {
26          int pi = partition(arr, low, high);
27
28          quickSort(arr, low, pi - 1);
29          quickSort(arr, pi + 1, high);
30      }
31  }
32
33  int main() {
34      int n;
35
36      scanf("%d", &n);
37
38      int arr[n];
39
40      for (int i = 0; i < n; i++) {
41          scanf("%d", &arr[i]);
42      }
43
44      quickSort(arr, 0, n - 1);
45
46      for (int i = 0; i < n; i++) {
47          if (i > 0)
```

```
48              printf(" ");
49          printf("%d", arr[i]);
50      }
51
52      return 0;
53  }
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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